

Akhilesh Babu Tumati

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PROFESSIONAL EXPERIENCE

Software Engineer, Tata Consultancy Services

Aug 2023 - Aug 2024 — Hyderabad, India

- Operated Jenkins CI/CD pipelines for UPS Client, achieving 40% faster deployment time and reducing critical deployment errors by 30% through automated testing and validation and Maintained CouchBase NoSQL database infrastructure, resulting in 25% revamp in data retrieval efficiency and 15% enhanced overall application performance
- Devised a data quality assurance framework that reduced data discrepancies by 40%, ensuring the accuracy and integrity of analytics outputs, which are critical for strategic planning and operational success and Completed comprehensive system monitoring using DynaTrace and Grafana, reducing system downtime by 20% and optimizing application response time by 10% through preemptive issue discovery

Software Developer, Honeywell

Jan 2023 - Jun 2023 — Hyderabad, India

- Prepared automated testing scripts using Katalon Studio, achieving 30% reduction in testing cycles and expanding test coverage by 25% across enterprise applications and Developed and enhanced frontend applications using JavaScript and Angular, delivering 20% faster page load times and increasing user satisfaction metrics by 15%
- Conducted Coverity and Blackduck security protocols, enhancing code quality metrics by 40% and strengthening application security standards and Streamlined application performance through Dynatrace monitoring, resulting in 35% increase in system efficiency and 20% reduction in critical system downtime

EDUCATION

Masters of Engineering in Computer Science

Aug 2024 - Present — Virginia, United States

Virginia Polytechnic Institute and State University: GPA: 4.0/4.0

Bachelor of Technology in Computer Science and Engineering

Jul 2019 - Aug 2023 — Andhra Pradesh, India

Vellore Institute of Technology: GPA: 3.5/4.0

SKILLS

Programming Languages: Java, JavaScript, Python, SQL, Maven, XML, Hibernate, JUnit

Software Development: TypeScript, React, Next.js, Node.js, Spring Boot, Sanity CMS, Katalon, Selenium, Jira, DynaTrace, Grafana, Coverity, Jenkins CI/CD, Bamboo, BitBucket, Git, Confluence, AWS (EC2, VPC, EKS, Lambda)

Databases: MySQL, CouchBase, MongoDB, Oracle SQL Developer

Data Science: NumPy, Pandas, Matplotlib, TensorFlow, scikit-learn, PyTorch, PowerBI, Keras

PROJECTS

Technoverse Blog,

Nov 2024 - Dec 2024

- Built a robust blog platform that reached outstanding performance on Vercel deployment using TypeScript and Next.js; employed Next.js alongside Sanity CMS for optimal user experience and comprehensive content management capabilities enhancing overall site efficiency.

Mahabharat Character Detection,

Aug 2022 - Dec 2022

- Conducted multi-character detection system utilizing RoboFlow, PyTorch, TensorFlow, CNN, ViT, YOLO, and ResNet-50, VGG-19, and Vision Transformers, achieving 30% improved accuracy over baseline models. Implemented YOLOv5 architecture for real-time character detection and classification in video streams.

Crypto Currency Value Prediction,

Sep 2021 - Nov 2021

- Created advanced time series prediction models using Google Colab, Python, SKLearn, Numpy, Pandas, Seaborn, and ARIMA and SARIMA algorithms, achieving 85% accuracy in cryptocurrency value forecasting. Developed interactive visualization dashboard using Streamlit for real-time forecast analysis and trend monitoring.

Creating Private Cloud Storage:

May 2021 - Jul 2021

- Architected scalable private cloud infrastructure using Raspberry Pi, NextCloud, Python, and HardDiskDrive for storage. Scheduled secure storage protocols and streamlined data transfer mechanisms for enhanced performance.

PUBLICATIONS

[1] Tumati, A. B., Gangaraju, R., Mannepli, B. R., Sumathi, D., & Alluri, B. K. (2023, January). Face Invariant Classification and Detection of Mythology Characters Using Custom Dataset (ClaDeMuC-CD). In *2023 Third International Conference on Advances in Electrical, Computing, Communication and Sustainable Technologies (ICAECT)* (pp. 1-8). IEEE. 15 May 2023

[2] Natarajan, K., Gangaraju, R., & Tumati, A. B. (2022). Hybrid ML and DL models for flood level prediction. *International Journal of Health Sciences*. 08 Dec 2022